

11 Show the symbol

Finally, you need to create the function that handles the button presses. This function will always display a symbol, but how it operates depends on whether it's the first or second turn in the matching attempt. If it's the first turn, the function just needs to remember which button was pressed. If it's the second turn, it needs to check if the symbols match. Symbols that don't match are hidden. Matching symbols are left showing and their buttons are disabled.

```
from tkinter import Tk, Button, DISABLED
```

```
def show_symbol(x, y):
```

```
    global first
```

```
    global previousX, previousY
```

```
    buttons[x, y]['text'] = button_symbols[x, y]
```

```
    buttons[x, y].update_idletasks()
```

```
    if first:
```

```
        previousX = x
```

```
        previousY = y
```

```
        first = False
```

```
    elif previousX != x or previousY != y:
```

```
        if buttons[previousX, previousY]['text'] != buttons[x, y]['text']:
```

```
            time.sleep(0.5)
```

```
            buttons[previousX, previousY]['text'] = ''
```

```
            buttons[x, y]['text'] = ''
```

```
        else:
```

```
            buttons[previousX, previousY]['command'] = DISABLED
```

```
            buttons[x, y]['command'] = DISABLED
```

```
        first = True
```

The **x** and **y** values tell the function which button has been pressed.

These lines tell the program that the variables are global.

These lines show the symbol.

If it's the first turn, the code remembers the button press by storing the **x** and **y** coordinates.

Second turn. This line includes a check to stop the player cheating by pressing every button twice!

If the symbols don't match...

If the symbols match...

Disable the pair of matching buttons so the player can't press them again.

Wait 0.5 seconds to give the player time to see the symbols, then hide them.

This line gets the function ready for the first button press of the next attempt.

△ How it works

The function shows a button's symbol by changing its text label to the Unicode character we randomly assigned to it. We use `update_idletasks()` to tell **Tkinter** to show this symbol right now. If it's the first turn, we just store the button's coordinates in variables. If it's the second turn, we need to check the player isn't trying to cheat by hitting the same button twice. If they aren't, we check if the symbols match. If the symbols don't match, we hide them by setting the text to empty strings; if they do match, we leave them showing but disable the buttons.

